E-JEPA-TTC: Cache-Free High-Resolution Event Prediction for Auditable TTC Research

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Reproducible local report from the `e-jepa-ttc` repository

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Abstract

E-JEPA-TTC investigates time-to-contact estimation from event cameras using high-resolution dense tokens and multi-horizon joint-embedding prediction. This revision does not claim state of the art. It combines an exact historical EvTTC baseline reconstruction, grouped cross-validation, an audit of the public Garl-TTC/eAP contracts, and a raw on-demand trainer that avoids a projected 455 GiB dense cache. It also adds a dataset-free, falsifiable audit of semantic shortcut allocation. The historical baseline obtains 0.322892 s MAE and 8.1554% relative error on its original validation split. Dense Patch improves one matched split but loses five-fold, three-seed grouped cross-validation to the global A0 control. A first real high-resolution 16/16-sample smoke completes the new pipeline but obtains sequence-macro MiD 1868.3186; it is integration evidence, not evidence of accuracy. The current bottleneck is therefore the learned representation—a compatible dense JEPA pretrainer and RGB-event fusion are still missing—rather than storage or basic pipeline execution.

1 Scope and claim boundary

The system targets signed continuous TTC from asynchronous events and later an RGB-event multimodal extension. EvTTC [1] supplies local supervised evaluation, whereas Garl-TTC/eAP [2] defines the principal external comparison. Event-only and RGB-E are separate protocols.

A state-of-the-art claim requires matching modality, split, metric, and benchmark; three predeclared seeds; a frozen checkpoint; label-free EvTTC prediction; and an official eAP/CodaBench result. None of these external gates is currently closed.

2 Historical EvTTC evidence

The historical EventTubelet baseline was reconstructed byte-for-byte on its original cache. Table 1 separates that anchor from the later matched architecture matrix.

Grouped-CV values are means across five folds and three seeds; their RTE standard deviations are 0.52 and 0.06 percentage points, respectively. A1 improves aggregate MAE by only 0.41%, worsens score and RTE, and costs approximately 2.16× inference latency. Attention Residuals, Object-KDA inspired by delta-memory designs [3], and bbox-ROI pooling also fail their sequential gates. These negative results prevent complexity from being reintroduced without a new predeclared hypothesis.

Table 1: Audited historical and grouped-CV evidence.

Model/protocol	Score	RTE	MAE (s)
BASE historical	–	8.1554%	0.3229
A0 grouped CV	0.58452	30.25%	1.011
A1 Dense grouped CV	0.59312	30.55%	1.007

3 Data and leakage controls

EvTTC-32 uses indivisible sequence groups; no random window split is allowed. The sealed EvTTC benchmark remains unopened. The local eAP inventory covers 40 of 46 expected sequences. A signed pilot uses 9/3 train/validation sequences and the full local split uses 32/8, selected without EvTTC metrics.

The Garl join, timestamps, calibration, signed TTC metrics, and balanced sampling were audited. TTC, depth, 3-D height, category, and masks are supervision or audit metadata, never event-encoder inputs. CARLA was removed from the active inventory after negative cross-domain transfer; compact negative summaries remain tracked.

4 High-resolution model

The active event-only path is

$$\begin{aligned} X &\to \text{patches} \to \text{window attention} \\ &\to \text{temporal mixer} \to \hat{\text{TTC}}. \end{aligned} \tag{1}$$

Padding and validity masks preserve non-divisible borders. Optional 2×2 space-to-depth merging reduces token count before the block-causal temporal mixer, which operates on $[B, T, P, D]$. Fixed learned queries pool retained patches for the TTC head. A memory guard estimates attention allocations before execution. A BF16

Table 2: Synthetic semantic-shortcut audit, three-seed mean.

Objective	Dyn. R^2	MAE	Acc.	Dup.
Repository variance	0.15	0.39	0.84	1.93
VISReg-style	0.20	0.38	0.92	1.63
Temporal residual	**0.72**	**0.29**	0.65	1.06
R^2 rate+dependence	0.29	0.36	0.88	1.92
Residual+R^2	0.48	0.34	0.68	**0.68**

query-pooling dtype defect found during the audit is now regression-tested.

The target is signed. Training minimizes Smooth L1 after

$$f(x) = \mathrm{sign}(x)\log(1 + |x|), \tag{2}$$

so separating objects are not incorrectly mapped to positive TTC. Model selection uses sequence-macro MiD rather than a window-weighted average.

5 JEPA status

The intended objective predicts future dense target-encoder tokens at multiple horizons, with stop-gradient and EMA. It must not use TTC or future labels. The existing pooled legacy pretrainer is architecture-incompatible with the active high-resolution token grid, so its alias now fails explicitly. The current high-resolution candidate consequently starts from random initialization unless a strictly compatible checkpoint is supplied.

This missing matched pretraining path is the principal scientific limitation. A latent loss decrease in another architecture would not establish downstream TTC transfer.

6 Semantic capacity audit

The real 192-dimensional SSL artifact has effective ranks 2.255, 1.095, and 5.105 for context, predictor, and target. The predictor is nearly one-dimensional, but rank cannot identify whether that axis represents TTC, expansion, or sequence identity. Healthy variance is not a semantic certificate, consistent with the known slow-feature failure of predictive embeddings [4].

We therefore added a dataset-free falsifier. Five representation objectives use identical observations and seeds 7/13/23; TTC and shortcut labels are withheld during representation training and used only by probes fitted after freezing. Table 2 reports the fixed-per-sequence 12-bit shortcut.

Variance and the VISReg-style distributional control [5] retain the slow shortcut. R^2-lite does not pass the predeclared TTC gate and is rejected for production. Temporal residual prediction passes the slow-nuisance gate, but in a frame-varying control its dynamic R^2 falls to -0.05, versus 0.74 for the repository objective. It is therefore conditional rather than universal. The minimal real candidate keeps a level channel for scale/content and adds a residual channel for expansion/motion; it does not introduce rate, HSIC, or conditional-MI estimators before a matched eAP screen.

Table 3: Raw event-only integration smoke; not promotable.

Quantity	Value
Train / validation samples	16 / 16
Epochs / seed	1 / 7
Elapsed time	17.16 s
Sequence-macro MiD	1868.3186
Global MiD	1941.2832
Weighted RTE	119.2892%
Failure rate	0%

7 Cache-free training

The trainer indexes raw HDF5 windows and materializes only the active batch. It implements deterministic balanced sampling, BF16, gradient accumulation, clipping, epoch-derived shuffle for reproducible resume, macro validation, and atomic best/ last checkpoints with provenance.

The screen uses dimension 32 and patch size 8. The full candidate uses dimension 192, patch size 16, batch 4 with accumulation 6, up to 30 epochs, and seeds 7/13/23. Full runs require a clean Git tree and share commit, configuration, data, and split hashes before freezing.

7.1 Storage audit

A full dense cache was projected at approximately 455 GiB. A 256-sample shard was materialized, SHA-verified, consumed, and removed. A 4096-sample attempt approached 11 GiB RAM without completing. The global-cache route is therefore excluded.

On 2 August, the rejected CARLA extraction (71.64 GiB), generated runs (16.87 GiB), and feature caches (12.81 GiB) were deleted. Compact signed summaries were retained. This recovered approximately 101 GiB in that cleanup pass.

8 Current real smoke

The zero failure rate shows that the poor metric is not hidden exception filtering. The run validates data flow, GPU execution, signed loss, metrics, and checkpoint compatibility only. Scaling this formulation immediately would be expensive and scientifically unjustified.

9 Evaluation orchestration

One runner separates the stages:

```
train -> freeze -> evttc-predict
     -> evttc-score -> submission-validate
```

The full freeze selects among seeds only on Garl validation. EvTTC prediction requires a hashed label-free manifest and rejects target-like fields before loading the checkpoint or GPU. Scoring receives targets in a separate process. Submission validation is offline and never counts as an official upload.

10 Limitations and next gates

Dense JEPA pretraining and RGB-E fusion remain unimplemented. The semantic audit is synthetic and does not prove real eAP shortcut allocation. Six eAP sequences, the real EvTTC Table-VI input manifest, full three-seed training, robustness, calibration, final ONNX export, and a real demo are missing. Causal bbox-free expansion/FoE has not surpassed A0, and ground-truth bbox/depth remain diagnostic oracles. The system is not suitable for safety control.

The next economical sequence is: (1) overfit matched dense JEPA with level and residual channels on 256 samples; (2) paired level-versus-level+residual screens with frozen TTC, expansion, event-rate, and sequence-ID probes; (3) isolated RGB-E; (4) real EvTTC predict/score; and only then (5) full 7/13/23 and official eAP/CodaBench submission. These restrictions preclude a present SOTA claim.

References

[1] J. Li et al. "EvTTC: An Event Camera Dataset for Time-to-Collision Estimation." arXiv:2412.05053, 2024.

[2] J. Li et al. "Toward Deep Representation Learning for Event-Enhanced Visual Autonomous Perception: the eAP Dataset." arXiv:2603.16303, 2026.

[3] Kimi Team. "Kimi K3: Open Frontier Intelligence." arXiv:2607.24653, 2026.

[4] V. Sobal et al. "Joint Embedding Predictive Architectures Focus on Slow Features." arXiv:2211.10831, 2022.

[5] "VISReg: Variance-Invariance-Sketching Regularization for JEPA Training." arXiv:2606.02572, 2026.